

One-Day-Ahead 1% VaR Forecasting on QQQ Daily Log Returns

A class report on rolling VaR comparison, backtesting, and forecast uncertainty

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1 Introduction

Value-at-Risk (VaR) is a standard summary of downside market risk. A one-day-ahead 1% VaR forecast gives the conditional lower-tail threshold for tomorrow's return such that only about 1% of outcomes are expected to fall below it, given information available today. This report studies that question for QQQ daily log returns using a rolling out-of-sample forecasting design.

The project compares several classes of VaR models: parametric volatility-based methods, semi-parametric empirical-tail methods, a nonparametric benchmark, a direct quantile model, and an exogenous-volatility benchmark using VIX. The empirical results indicate that the most effective models in this application are those that combine dynamic volatility estimation with empirical tail calibration. In particular, **GJR-FHS** emerges as the strongest overall specification, with **GARCH-FHS** a close second. Historical Simulation remains a strong benchmark, whereas Gaussian-tail specifications tend to underestimate left-tail risk.

The report is written for classmates and has two purposes. First, it explains the main ideas behind one-day-ahead VaR forecasting and the distinctions among the competing methods. Second, it uses the saved project outputs to evaluate which modeling choices perform best in this specific QQQ application.

2 Historical Background

Modern VaR became especially prominent during the 1990s, when financial institutions sought a compact measure of market risk over a fixed horizon and confidence level. A major practical milestone was J.P. Morgan’s *RiskMetrics* framework, which popularized variance-based VaR measurement and helped standardize daily risk reporting. The 1996 Basel amendment then gave VaR regulatory significance by allowing banks to use internal risk models, subject to formal backtesting requirements.

The econometric foundations of VaR forecasting are closely tied to volatility modeling. Engle’s ARCH model and Bollerslev’s GARCH extension established a framework for capturing volatility clustering, one of the most persistent empirical features of financial returns. Even when returns themselves exhibit little serial dependence, their conditional variance is often highly persistent, making dynamic volatility forecasts a natural basis for conditional tail-risk measurement.

As VaR became more widely used, attention turned from construction to validation. Kupiec (1995) proposed the unconditional coverage test, which examines whether the observed frequency of VaR exceedances matches the nominal level. Christoffersen (1998) extended this logic by emphasizing independence: a satisfactory VaR model should not only produce the correct number of violations on average, but should also avoid clustering those violations through time. These tests remain standard tools in empirical VaR evaluation.

A further development was the move away from fully parametric tail assumptions. Barone-Adesi, Giannopoulos, and Vosper (1999) developed Filtered Historical Simulation (FHS), which combines a parametric volatility forecast with the empirical distribution of standardized residuals. This hybrid approach is particularly attractive in financial applications, where volatility dynamics may be captured reasonably well by GARCH-type models, while the innovation distribution often exhibits heavier tails than the Gaussian benchmark allows.

A different branch of the literature models the quantile directly rather than modeling the full conditional distribution first. Engle and Manganelli’s CAViaR framework is the leading example. In that setting, the conditional quantile itself evolves dynamically and is estimated by minimizing a quantile-regression objective. This is theoretically appealing, though in practice it requires careful optimization and dedicated adequacy testing.

Finally, there has been sustained interest in extending conditional variance models to include exogenous variables. In this project, that role is played by VIX through a GARCHX specification. This benchmark asks whether implied market volatility adds incremental information about next-day QQQ downside risk beyond what is already contained in lagged returns.

3 Purpose Statement and Mathematical Development

3.1 Purpose

Let r_t denote the QQQ daily log return,

$$r_t = \log \left(\frac{P_t}{P_{t-1}} \right),$$

where P_t is the adjusted closing price. The one-day-ahead 1% VaR is the conditional lower-tail quantile satisfying

$$\Pr(r_{t+1} < \text{VaR}_{t+1}(0.01) \mid \mathcal{F}_t) = 0.01.$$

The project is concerned with return-space forecasting, so the object of interest is the conditional 1% quantile of r_{t+1} rather than a loss measure in dollar units. The practical objective is not in-sample fit, but out-of-sample calibration: which model most reliably forecasts next-day downside risk under a realistic rolling exercise?

3.2 Rolling forecasting design

The saved project pipeline uses 2,824 complete QQQ/VIX observations after cleaning, spanning 2015-01-05 to 2026-03-27. The initial training fraction is 70%, giving:

- initial training size: 1,976 observations,
- out-of-sample size: 848 one-day forecasts,
- first forecast date: 2022-11-08,
- expected number of 1% violations: 8.48.

At each out-of-sample date t , the procedure is:

1. estimate the model using information available through day $t - 1$,
2. produce a one-day-ahead VaR forecast for day t ,
3. observe the realized return r_t ,
4. record the hit indicator

$$I_t = \mathbf{1}\{r_t < \widehat{\text{VaR}}_t\},$$

5. evaluate the resulting hit sequence with standard backtests.

This forecasting design is appropriate because VaR is a risk-management tool. The relevant question is whether a model remains calibrated when used sequentially in real time, not whether it explains the historical sample retrospectively.

3.3 Common location-scale structure

Several of the models can be written in conditional location-scale form,

$$r_{t+1} = \mu_{t+1} + \sigma_{t+1}z_{t+1},$$

which implies

$$\text{VaR}_{t+1}(\alpha) = \mu_{t+1} + \sigma_{t+1}q_\alpha.$$

The comparison therefore reduces to two central modeling questions:

- how the conditional scale σ_{t+1} is forecast, and

- how the tail quantile q_α is determined.

This decomposition provides a useful way to organize the competing methods.

3.4 Models used in the project

3.4.1 GARCH(1,1): parametric volatility baseline

The standard GARCH(1,1) model takes

$$\varepsilon_t = r_t - \mu,$$

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2.$$

In the saved implementation, the main GARCH baseline uses Student- t innovations. The corresponding one-step VaR forecast is

$$\widehat{\text{VaR}}_{t+1}(\alpha) = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1} q_{\alpha,t}^{(t)},$$

where $q_{\alpha,t}^{(t)}$ is the fitted standardized Student- t quantile. This serves as a transparent parametric benchmark: volatility is dynamic, but tail behavior is still imposed through a parametric distribution.

3.4.2 GJR-GARCH: leverage-sensitive parametric baseline

To allow asymmetry in volatility responses, GJR-GARCH uses

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \gamma \varepsilon_{t-1}^2 I(\varepsilon_{t-1} < 0) + \beta \sigma_{t-1}^2.$$

When $\gamma > 0$, negative return shocks raise future volatility more than positive shocks of the same magnitude. This feature is empirically plausible for equity returns and makes GJR-GARCH a natural extension of the symmetric GARCH baseline.

3.4.3 Historical Simulation: nonparametric benchmark

Historical Simulation uses the empirical lower-tail quantile of the most recent W returns,

$$\widehat{\text{VaR}}_{t+1}(\alpha) = \text{Quantile}_\alpha(r_{t-W+1}, \dots, r_t).$$

In the saved project, $W = 250$. This method is transparent and assumption-light, though it can adjust slowly when volatility changes abruptly.

3.4.4 Filtered Historical Simulation: semi-parametric empirical-tail VaR

Filtered Historical Simulation first estimates a volatility model and then applies the empirical lower-tail quantile of the standardized residuals:

$$\widehat{\text{VaR}}_{t+1}(\alpha) = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1} \hat{q}_{\alpha, \text{emp}}.$$

This is the central methodological result of the project. Volatility dynamics are modeled parametrically, but the innovation tail is learned from the residual distribution rather than imposed through a thin Gaussian shape. The saved workflow contains both Gaussian-based and Student- t -based FHS variants:

- GARCH-FHS,
- GJR-FHS,
- GARCH- t -FHS,
- GJR- t -FHS.

3.4.5 Residual bootstrap VaR

The bootstrap variants resample standardized residuals, simulate one-step returns, and estimate VaR as the empirical 1% quantile of the simulated distribution:

$$r_{t+1}^{*(i)} = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1} z_{t+1}^{*(i)}.$$

The VaR estimate is then the empirical α -quantile across the simulated returns. For a one-step forecast horizon, this is mechanically close to FHS, which helps explain why the empirical results are similar.

3.4.6 CAViaR-SAV: direct quantile dynamics

The implemented CAViaR specification is the Symmetric Absolute Value (SAV) form. The code evolves the positive VaR magnitude by

$$v_t = \beta_0 + \beta_1 v_{t-1} + \beta_2 |r_{t-1}|,$$

and maps back to the left-tail quantile through

$$q_t = -v_t.$$

Parameters are estimated by minimizing the quantile check-loss function

$$L(\beta) = \sum_t \rho_\alpha(r_t - q_t(\beta)),$$

where

$$\rho_\alpha(u) = u(\alpha - I(u < 0)).$$

Conceptually, this is attractive because it models the quantile directly. In the present project, however, it should be interpreted as exploratory rather than fully developed: only the SAV form is implemented, optimization is limited, and Dynamic Quantile-style adequacy testing is not included.

3.4.7 RiskMetrics: historically important benchmark

RiskMetrics uses the exponentially weighted recursion

$$\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) r_{t-1}^2,$$

with $\lambda = 0.94$ in the saved benchmark. It is historically important and easy to explain, but it relies on a Gaussian tail mapping and is therefore expected to struggle when downside tails are heavier than Gaussian theory suggests.

3.4.8 GARCHX(VIX): exogenous-volatility benchmark

The separate R benchmark uses the `garchx` package. In each rolling window, returns are demeaned and VIX is standardized, giving the conditional variance structure

$$y_t = r_t - \mu,$$

$$\sigma_t^2 = \omega + \alpha y_{t-1}^2 + \beta \sigma_{t-1}^2 + \delta x_t,$$

where x_t is the standardized VIX regressor. The current script converts the volatility forecast into VaR through a Gaussian quantile:

$$\widehat{\text{VaR}}_{t+1}(\alpha) = \mu + \hat{\sigma}_{t+1|t} \Phi^{-1}(\alpha).$$

This is best interpreted as a package-based Gaussian-QML GARCHX comparison benchmark. It is cleaner than the earlier custom implementation, but it remains a parametric-tail method rather than an empirical-tail specification.

4 Data Example with Representative Code

4.1 Data setting and forecasting context

The project uses QQQ adjusted closing prices and VIX closing values. QQQ daily log returns are the target series, while VIX enters only through the GARCHX benchmark. The rolling design produces a sequence of one-step-ahead forecasts, each based only on information available at the forecast origin.

A representative forecast date is 2025-04-04, a day with a large negative realized return. This date is informative because it highlights how the models differ when market stress is nontrivial.

Date	Model	Forecasted VaR	Realized return	Violation?
2025-04-04	GJR-FHS	-0.07956	-0.06412	0
2025-04-04	GARCH-FHS	-0.07059	-0.06412	0
2025-04-04	GARCH(1,1)	-0.06311	-0.06412	1
2025-04-04	Historical-Simulation	-0.03664	-0.06412	1
2025-04-04	RiskMetrics ($\lambda = 0.94$)	-0.04525	-0.06412	1
2025-04-04	GARCHX(VIX) [garchx pkg]	-0.04914	-0.06412	1
2025-04-04	CAViaR-SAV	-0.06293	-0.06412	1

This single date does not determine the final ranking, but it makes the model differences concrete. The leading FHS specifications move materially farther into the left tail, while several benchmarks remain too close to zero and are breached.

4.2 Representative model-side code

The report emphasizes code that illustrates how forecasts are formed, rather than data-import or file-handling routines. The following examples are representative of the actual model logic used in the project.

4.2.1 GARCH one-step VaR mapping

```
fit = arch_model(r_train * 100, p=1, q=1, dist="StudentsT").fit(dis="off")
mu_hat = fit.params.get("mu", 0.0) / 100
sigma_hat = np.sqrt(fit.forecast(horizon=1, reindex=False).variance.values[-1, 0]) / 100
nu_hat = fit.params["nu"]
q_alpha = t.ppf(alpha, df=nu_hat) * np.sqrt((nu_hat - 2) / nu_hat)
var_t1 = mu_hat + sigma_hat * q_alpha
```

This code shows the essential parametric construction: estimate conditional volatility, obtain the fitted innovation quantile, and combine the two to form the next-day VaR.

4.2.2 Filtered Historical Simulation

```
fit = arch_model(r_train * 100, p=1, o=1, q=1, dist="normal").fit(disp="off")
mu_hat = fit.params.get("mu", 0.0) / 100
sigma_hat = np.sqrt(fit.forecast(horizon=1, reindex=False).variance.values[-1, 0]) / 100
z_hat = fit.std_resid.dropna().to_numpy()
q_emp = np.quantile(z_hat, alpha)
var_t1 = mu_hat + sigma_hat * q_emp
```

This is the key empirical-tail idea in the project: forecast volatility parametrically, but estimate the 1% tail from the empirical residual distribution.

4.2.3 Residual bootstrap VaR

```
z_star = np.random.choice(z_hat, size=B, replace=True)
r_star = mu_hat + sigma_hat * z_star
var_t1 = np.quantile(r_star, alpha)
```

For one-step forecasting, the bootstrap approach is very close in spirit to FHS, because both recover the predictive tail from standardized residuals rather than from a fixed parametric distribution.

4.2.4 RiskMetrics recursion

```
sigma2 = np.var(r_train[-250:])
for x in r_train[-250:]:
    sigma2 = lam * sigma2 + (1 - lam) * x**2
var_t1 = np.sqrt(sigma2) * norm.ppf(alpha)
```

This benchmark is included because it is historically influential and easy to interpret, even though it imposes a Gaussian lower tail.

4.2.5 CAViaR-SAV update and estimation objective

```
def sav_path(beta, r, v0):
    v = np.empty(len(r))
    v[0] = v0
    for t in range(1, len(r)):
        v[t] = beta[0] + beta[1] * v[t-1] + beta[2] * abs(r[t-1])
    return -v

loss = np.sum((alpha - (r < q_path).astype(float)) * (r - q_path))
```

This snippet illustrates the direct-quantile logic of CAViaR-SAV: the quantile itself evolves recursively and is estimated via the check-loss criterion.

4.3 Brief provenance clarification

The final comparison is assembled from multiple saved forecast files rather than a single unified runner. The main Python runner, the extension runner, the RiskMetrics benchmark, and the package-based GARCHX benchmark all share the same out-of-sample length of 848 days, which makes the combined comparison substantively coherent. That provenance matters for transparency, but the focus of the report remains the econometric comparison rather than the repository workflow.

5 Empirical Results

5.1 Backtest comparison

Table 2 summarizes the one-day-ahead 1% VaR backtests across the saved outputs. Since all reported models are evaluated on the same 848 out-of-sample observations, the comparison is directly interpretable.

Model	Source	Violations	Rate	Kupiec p	Christof- fersen p	Conditional coverage p
GJR-FHS	extension runner	8	0.94%	0.8672	0.6961	0.9137
GARCH-FHS	extension runner	7	0.83%	0.5984	0.7327	0.8212
GARCH-t-FHS	main runner	6	0.71%	0.3664	0.7698	0.6371
GJR-t-FHS	main runner	6	0.71%	0.3664	0.7698	0.6371
GARCH-t-Bootstrap	main runner	6	0.71%	0.3664	0.7698	0.6371
Historical-Simulation	main runner	12	1.42%	0.2529	0.1569	0.1910
GARCH(1,1)	main runner	13	1.53%	0.1480	0.1890	0.1482
CAViaR-SAV	main runner	6	0.71%	0.3664	0.0297	0.0625
GJR-GARCH	main runner	15	1.77%	0.0424	0.4621	0.0972
GARCHX(VRX) [garchx pkg]	VRX ench- mark	15	1.77%	0.0424	0.2621	0.0679
GARCH-Bootstrap	extension runner	5	0.59%	0.1934	0.8075	0.4166

Model	Source	Violations	Rate	Kupiec p	Christof- fersen p	Conditional coverage p
RiskMet- rics ($\lambda =$ 0.94)	RiskMet- rics bench- mark	18	2.12%	0.0043	0.3925	0.0117
GARCH- Gaussian	extension runner	22	2.59%	0.0001	0.5970	0.0005

Several patterns are especially clear.

5.1.1 1. The strongest overall evidence favors GJR-FHS and GARCH-FHS

Among the saved outputs, **GJR-FHS** is closest to the nominal 1% violation rate and passes unconditional coverage, independence, and conditional coverage quite comfortably. **GARCH-FHS** performs almost as well and is only slightly more conservative. The backtest evidence therefore supports these two specifications as the strongest practical models in the comparison.

5.1.2 2. Empirical-tail calibration is more important than simply changing the volatility recursion

The Student- t empirical-tail variants also perform well, and this should be acknowledged directly. However, the implemented comparison shows that the main gain comes from empirical-tail calibration itself. Simply replacing GARCH with GJR-GARCH without adjusting the tail treatment does not solve the left-tail problem. By contrast, once the residual distribution is used empirically, calibration improves substantially.

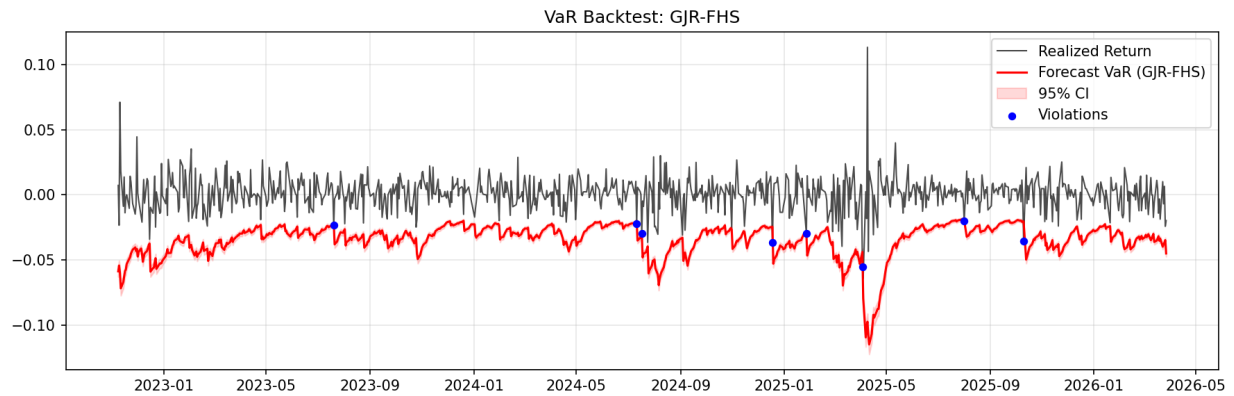
5.1.3 3. Gaussian tails are too thin for this application

RiskMetrics and GARCH-Gaussian both generate too many violations. The empirical evidence suggests that the issue is not the idea of dynamic volatility forecasting per se, but the inadequacy of a Gaussian lower-tail mapping for QQQ downside risk. In this sample, the left tail is too heavy for a Gaussian quantile to provide a reliable 1% cutoff.

5.2 Evidence from the plots

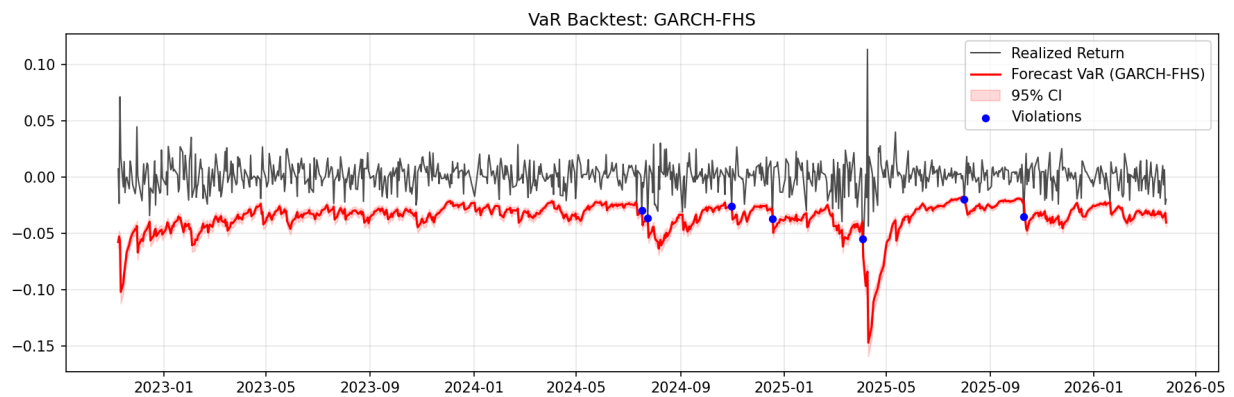
The saved VaR plots complement the backtest statistics by showing how the competing methods behave during periods of market stress.

5.2.1 Best overall model: GJR-FHS



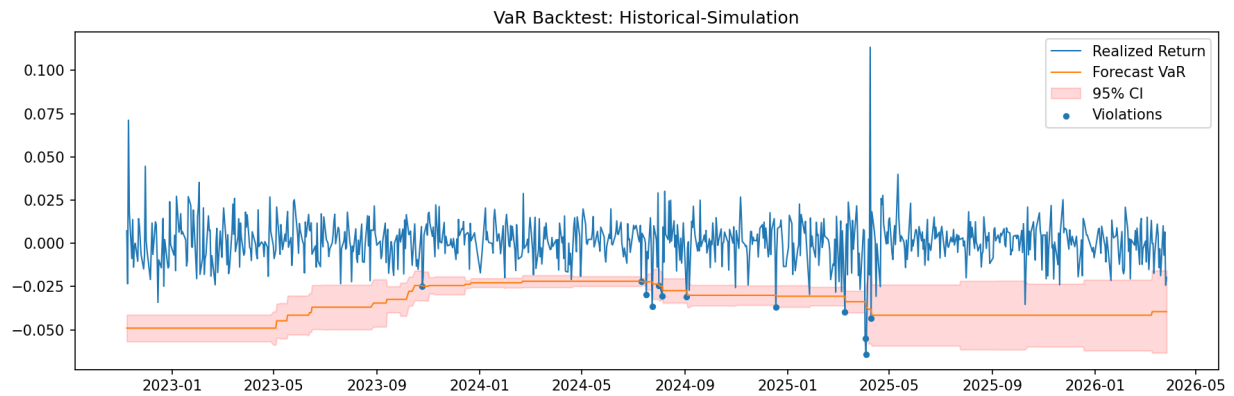
The GJR-FHS VaR line moves sharply downward during volatile episodes while remaining reasonably adaptive once stress subsides. This visual behavior is consistent with its strong backtest performance.

5.2.2 Close second: GARCH-FHS



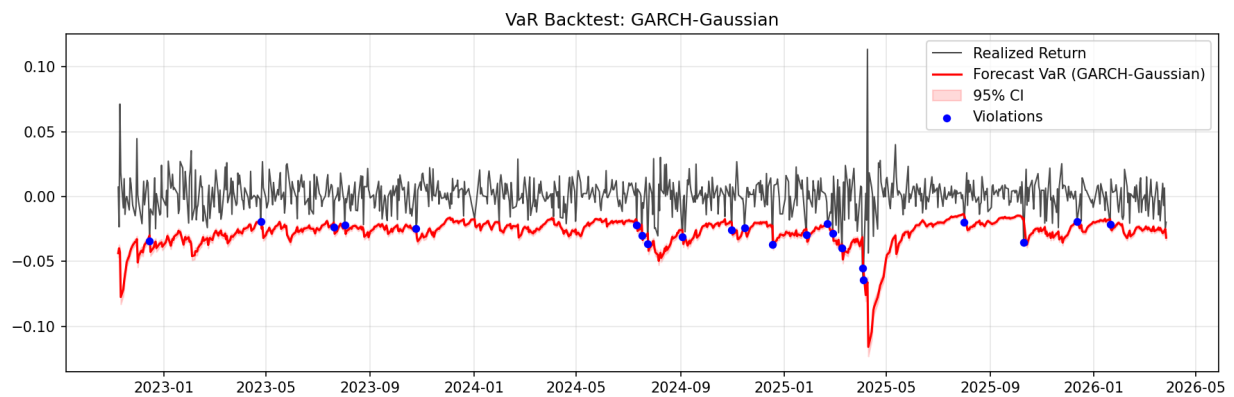
GARCH-FHS exhibits a very similar pattern. Relative to GJR-FHS, it is somewhat less flexible with respect to asymmetry, but its empirical calibration remains excellent.

5.2.3 Strong benchmark: Historical Simulation



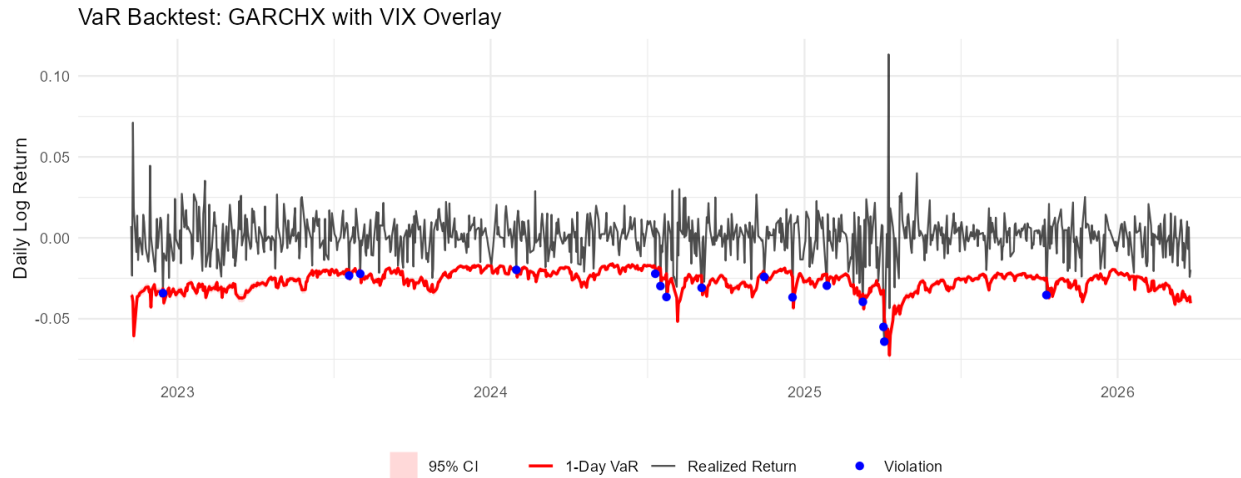
Historical Simulation is more stepwise and adjusts more slowly. That limitation is visible during turbulent periods, yet its overall performance remains respectable, which helps explain why it remains a useful benchmark.

5.2.4 Why Gaussian models underperform



The GARCH-Gaussian VaR line remains too close to zero in stressed periods. The plot therefore makes the backtest outcome intuitive: the Gaussian tail does not extend far enough into the lower tail of the return distribution.

5.2.5 Exogenous-volatility benchmark



The package-based GARCHX benchmark produces a coherent volatility forecast, but its Gaussian tail conversion remains too shallow relative to the best empirical-tail specifications. It is therefore useful as a comparison benchmark, but not as the preferred forecasting model.

5.3 Final model positioning

The empirical comparison supports the following substantive positioning:

- **main recommendation:** GJR-FHS,
- **close second:** GARCH-FHS,
- **strong benchmark:** Historical Simulation,
- **useful parametric baseline:** GARCH(1,1),
- **comparison benchmark with exogenous information:** package-based GARCHX(VIX),
- **exploratory direct-quantile model:** CAViaR-SAV.

The Student- t empirical-tail variants also perform very well and should not be ignored. Even so, the most parsimonious interpretation of the full comparison is that GJR-FHS and GARCH-FHS provide the clearest combination of statistical performance, interpretability, and methodological coherence.

6 Forecast Uncertainty and Confidence Interval Interpretation

The assignment requires some discussion of forecast uncertainty, and the saved forecast files provide this through reported lower and upper bounds for each VaR forecast. These bounds make it possible to examine uncertainty in the estimated conditional quantile rather than considering only point forecasts.

6.1 Why uncertainty matters

A VaR forecast is itself an estimate, so it inherits uncertainty from several sources:

- finite-sample estimation error,
- uncertainty in volatility estimation,
- uncertainty in the tail quantile,
- model misspecification.

A generic large-sample quantile interval can be written as

$$\hat{q}_\alpha \pm 1.96 \text{SE}(\hat{q}_\alpha),$$

where

$$\text{SE}(\hat{q}_\alpha) = \sqrt{\frac{\alpha(1-\alpha)}{n f(\hat{q}_\alpha)^2}}.$$

For location-scale VaR models,

$$\text{VaR}_{t+1}(\alpha) = \mu_{t+1} + \sigma_{t+1}q_\alpha,$$

so uncertainty in the tail quantile propagates directly into uncertainty in the VaR forecast.

6.2 Why interval widths are not perfectly comparable

An important qualification is that the reported intervals are not constructed identically across all model families.

- **Parametric Gaussian and Student- t models** use asymptotic quantile-standard-error logic mapped through the location-scale formula.
- **FHS models** combine parametric volatility estimation with an empirical residual quantile, so their intervals reflect uncertainty in that empirical tail estimate.
- **Bootstrap models** derive interval information from the simulated predictive distribution.
- **Historical Simulation** uses a rolling empirical-quantile interval from the raw returns.
- **CAViaR-SAV** uses a centered empirical proxy interval rather than a full inferential interval derived from a likelihood-based covariance calculation.

Accordingly, interval widths are informative, but they should be interpreted as model-specific measures of VaR uncertainty rather than perfectly homogeneous statistical objects.

6.3 Average interval widths from the saved forecasts

A simple summary is the average width

$$\text{width}_t = \text{upper_bound}_t - \text{lower_bound}_t.$$

Using the saved forecast files, the average widths are:

Model	Mean interval width
CAViaR-SAV	0.02064
Historical-Simulation	0.02064
GARCH-t-FHS	0.00783
GARCH-FHS	0.00688
GARCH(1,1)	0.00677
GJR-GARCH	0.00637
GJR-t-FHS	0.00586
GJR-FHS	0.00436
GARCH-Gaussian	0.00369
GARCH-t-Bootstrap	0.00368
RiskMetrics ($\lambda = 0.94$)	0.00368
GARCHX(VIX) [garchx pkg]	0.00353
GARCH-Bootstrap	0.00303

Several interpretations follow.

6.3.1 1. Historical Simulation and CAViaR-SAV are the least precise

These two models have the widest average intervals by a substantial margin. For Historical Simulation this is unsurprising, since an extreme empirical quantile estimated from a rolling 250-day window is inherently noisy. For CAViaR-SAV, the large width reflects both quantile uncertainty and the proxy nature of the interval construction in the current implementation.

6.3.2 2. The leading FHS models are both well calibrated and reasonably stable

GJR-FHS and GARCH-FHS are appealing not because they minimize interval width mechanically, but because they combine moderate interval width with strong backtesting performance. That joint behavior is more informative than width alone.

6.3.3 3. Narrow intervals are not necessarily an advantage when calibration is poor

GARCH-Gaussian, RiskMetrics, and the package-based GARCHX benchmark all have comparatively narrow intervals. Yet their backtest performance is weaker than that of the leading FHS models. A narrow interval around a biased or insufficiently deep VaR forecast is not, by itself, desirable.

6.3.4 4. Bootstrap widths should be interpreted with care

The bootstrap variants produce relatively small average widths. For one-step forecasting this is not surprising, since bootstrap and FHS are already closely related. In addition, the current bootstrap implementation is deterministic across rolling iterations because the seed is reset within the forecast function, so the resulting intervals should be interpreted cautiously.

6.4 Practical interpretation

A wider VaR interval indicates greater uncertainty about the location of the next-day 1% threshold; a narrower interval indicates a more concentrated estimate within that model's own construction. The backtest evidence and uncertainty analysis point in the same direction: the most convincing models are those that combine credible calibration with reasonably stable uncertainty behavior. In this project, that description fits **GJR-FHS** and **GARCH-FHS** best.

7 Discussion

The empirical results suggest that the main improvement in this QQQ application comes from **empirical-tail calibration**, not merely from a more elaborate volatility recursion. Dynamic volatility modeling remains essential, but accurate variance forecasts alone are not sufficient if the lower-tail distribution is misspecified.

Asymmetry helps at the margin, which helps explain why GJR-FHS slightly outperforms GARCH-FHS. However, the comparison also shows that asymmetry by itself is not decisive. The more important modeling choice is whether the left tail is learned empirically from standardized residuals or imposed parametrically.

Historical Simulation remains an instructive benchmark. It is simpler and slower to adjust than the leading FHS models, yet it performs well enough to remind us that additional complexity must be justified empirically. This is a useful comparison point in a class report, because it shows that sophisticated methods do not automatically dominate straightforward benchmarks.

The package-based GARCHX(VIX) benchmark is conceptually worthwhile because implied volatility is economically meaningful and could plausibly improve tail forecasting. In the present saved outputs, however, it does not outperform the best return-based empirical-tail models. The evidence therefore supports treating it as a useful comparison rather than a preferred specification.

The CAViaR-SAV result also warrants a balanced interpretation. The saved outputs do not support describing it as a failure of unconditional coverage. Rather, the model appears conservative in terms of hit frequency but weaker with respect to independence, and its current implementation remains limited by relatively light optimization and incomplete adequacy testing.

8 Limitations

Several limitations should be acknowledged explicitly.

1. **Split workflow.** The final comparison is assembled from multiple saved runners rather than a single unified estimation pipeline. The comparison remains meaningful because the out-of-sample horizon is aligned across models, but the workflow is less integrated than ideal.
2. **GARCHX benchmark scope.** The package-based GARCHX benchmark is cleaner than the earlier custom implementation, but it still uses a Gaussian VaR overlay. It should therefore be interpreted as a comparison benchmark rather than as a final recommended model.
3. **CAViaR implementation scope.** Only the SAV specification is implemented, optimization is limited, and Dynamic Quantile-style adequacy testing is absent. The result is informative, but it is not a full replication of the original CAViaR framework.
4. **Bootstrap implementation detail.** The current bootstrap code resets the random seed within the forecast function. This improves reproducibility, but it makes the rolling bootstrap mechanically deterministic.
5. **Application scope.** The conclusions are specific to one asset, one forecast horizon, and one tail probability. The ranking could differ for other assets, multi-day VaR, or related risk measures such as Expected Shortfall.

9 Conclusion

This report examined one-day-ahead 1% VaR forecasting for QQQ daily log returns using a rolling out-of-sample design with 848 forecast dates. The comparison across saved project outputs yields a consistent substantive conclusion.

The backtest evidence supports the view that the most effective models are those that combine **dynamic volatility estimation with empirical tail calibration**. The strongest practical recommendation is **GJR-FHS**, with **GARCH-FHS** a close second. Historical Simulation remains a strong benchmark, and GARCH(1,1) remains a useful parametric baseline. By contrast, Gaussian-tail specifications such as GARCH-Gaussian and RiskMetrics tend to underestimate downside risk in this application.

The package-based **GARCHX(VIX)** benchmark is worth including because it tests whether exogenous market volatility improves forecasting performance. In the saved outputs, however, it does not outperform the leading empirical-tail models. The exploratory **CAViaR-SAV** result is also worth reporting carefully: it is not primarily a coverage failure, but it remains limited by exception clustering and incomplete implementation.

The uncertainty analysis reinforces the same conclusion. Interval widths differ across model families and should not be treated as perfectly comparable, yet the most persuasive models are still those that combine reasonable uncertainty behavior with strong backtest calibration. On that criterion, **GJR-FHS** and **GARCH-FHS** remain the most convincing choices.

A parsimonious interpretation of the full project is therefore the following: for one-day-ahead 1% VaR on QQQ, the most successful approach is not simply to forecast volatility, but to forecast volatility dynamically **and** allow the tail to be learned more flexibly from the data.

10 References

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